

Master's Thesis
(working draft)

Circle Bundles with Positive Scalar
Curvature over Enlargeable Manifolds

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Introduction

The scalar curvature of a Riemannian Manifold (M, g) is a smooth function $\text{scal}: M \rightarrow \mathbb{R}$ measuring how much the volume of small geodesic balls around a point deviates from that of euclidian balls. While it is the weakest curvature condition for a Riemannian manifold, the presence of positive scalar curvature can still have strong topological implications. The archetypal example is the celebrated Gauß-Bonnet Theorem:

Theorem (Gauß-Bonnet). *Let (M, g) be a closed Riemannian 2-manifold. Then*

$$\int_M \text{scal} \, dA = 4\pi\chi(M),$$

where $\chi(M)$ is the Euler characteristic of M .

For non-compact complete manifolds, merely having positive scalar curvature may be too weak of a condition. The problem is that scal can still tend to zero quite rapidly at infinity. However, if the rate of decay at infinity can be controlled, positive scalar curvature still has profound consequences. For example, Chen has recently shown a decomposition result for 3-manifolds under the assumption that a positive scalar curvature metric with controlled decay at infinity exists [Chen2025]. This naturally raises the question of how fast the scalar curvature of a given complete Riemannian manifold X decays at infinity. One statement in this direction is the following theorem by Gromov:

Theorem (Enlargeable Decay Theorem, [Gro18], page 15f.). *Let X be a complete Riemannian manifold, $E \rightarrow M$ a two dimensional vector bundle over an enlargeable manifold M . Assume that*

- (i) *There exists a proper map $f: X \rightarrow E$, $\deg(f) \neq 0$,*
- (ii) *The total space of the circle bundle corresponding to E is also enlargeable.*

Then

$$\min_{B_{x_0}(R)} \text{scal}(X) \leq \frac{8\pi^2}{(R - R_0)^2}$$

for some $R_0 = R_0(X, x_0)$ and all $R > R_0$.

It is easy to see that condition (ii) is always fulfilled if the bundle $E \rightarrow M$ is trivial (compare Lemma 3.3 (i)), which lead Gromov to pose the following question:

Question ([Gro18]). “What happens to nontrivial bundles $E \rightarrow M$? [...] In general, the examples in [BH10] indicate a possibility of non-enlargeable circle bundles over enlargeable M ; yet, it seems hard(er) to find such examples, where the corresponding total space would admit metrics with $\text{scal} > 0$.”

As it turns out, such examples do in fact exist, that is, circle bundles over enlargeable manifolds, where the total space admits a metric of positive scalar curvature and thus, in particular, is not enlargeable. Bundles with this behaviour have recently been constructed in a paper by Kumar and Sen [KS25] by utilizing tools from symplectic geometry. A very rough sketch of the construction can be given as follows. Take M to be a so-called Donaldson divisor of a suitable closed symplectic manifold, e.g., the torus T^{2n} , $n \geq 3$. Such a Donaldson divisor is a certain codimension 2 symplectic submanifold $M^{2n-2} \subseteq T^{2n}$, which in some cases can be shown to be enlargeable. Now define a circle bundle $E := \partial\nu(M)$ over M as the boundary of a tubular neighbourhood $\nu(M) \subseteq T^{2n}$ of M . In order to construct a positive scalar curvature metric on E , consider the nullbordism $W := T^{2n} - \nu(M)$ obtained by removing the tubular neighbourhood from the ambient manifold. Now explicitly construct a Morse function $\psi: W \rightarrow \mathbb{R}$ with critical points of index at most n , which shows that E is obtained from the empty set by surgeries of codimension at least $n \geq 3$. Thus, E admits a metric of positive scalar curvature by the Gromov-Lawson-Schoen-Yau Surgery Theorem (see Theorem 2.1).

In this thesis, we provide an alternative, much more direct construction relying simply on basic topological concepts. It should be said, however, that this alternative approach only provides examples in the totally non-spin case.

The thesis is organized as follows. Chapter 1 recalls some concepts from the broader theory of principal bundles, laying the foundation for the subsequent chapters. In Chapter 2, we briefly introduce the reader to the theory of positive scalar curvature and prove two existence theorems for positive scalar curvature metrics, which are used in the main construction in Chapter 4. The proofs of these theorems are based on methods from surgery theory, which we also take some time to explain. Chapter 3 provides a brief discussion of the concept of enlargeability, which is the central idea underlying this thesis. Finally, Chapter 4 contains the main results. We present a simple construction of circle bundles over enlargeable manifolds whose total spaces admit metrics of positive scalar curvature in all dimensions ≥ 4 . These bundles are then used to obtain counterexamples to the Enlargeable Decay Theorem above if condition (ii) is dropped.

1 Preliminaries

This section develops the necessary preliminaries for what follows, by recalling the notions of principal bundles, classifying spaces and characteristic classes. We assume the reader to be familiar with the concept of a fiber bundle with structure group. A classical reference for this is for example [Ste51].

1.1 Principal Bundles and Associated Bundles

Definition 1.1. Let G be a topological group. A G -principal bundle is a fiber bundle with fiber G and structure group G acting by left translations.

The total space of a G -principal bundle $P \xrightarrow{\pi} B$ carries a natural right action of G defined as follows. Around $y \in P$ choose a local trivialisation

$$\phi: U \times G \rightarrow \pi^{-1}(U)$$

and let $\phi^{-1}(y) = (\pi(y), h)$. Then set

$$y \cdot g := \phi(\pi(y), hg).$$

This is well defined since by definition G acts on itself by *left* translations under a change of local trivialisation. A morphism of principal bundles should respect this right action, thus we define:

Definition 1.2. A morphism from a G -principal bundle $P \rightarrow B$ to an H -principal bundle $P' \rightarrow B'$ is a pair (F, f) of continuous maps $F: P \rightarrow P'$ and $f: B \rightarrow B'$ together with a homomorphism $\rho: G \rightarrow H$, such that

$$\begin{array}{ccc} P & \xrightarrow{F} & P' \\ \downarrow & & \downarrow \\ B & \xrightarrow{f} & B' \end{array}$$

commutes and such that F is ρ -equivariant, that is $F(x \cdot g) = F(x) \cdot \rho(g)$ for all $x \in P$ and $g \in G$. If $G = H$ we assume ρ to be the identity and if $B = B'$ we assume f to be the identity.

The condition of equivariance in the above definition is actually quite strong as illustrated by the following proposition.

Proposition 1.3. *Any morphism of G -principal bundles over the same base space is an isomorphism.*

Proof. First suppose we are given a morphism

$$F: B \times G \rightarrow B \times G$$

of trivial bundles. Then by definition F is of the form

$$F(x, g) = (x, \phi_x(g))$$

for a continuous family of equivariant maps $\phi_x: G \rightarrow G$. By equivariance $\phi_x(g) = \phi_x(e)g$ and thus an inverse of F is given by

$$F^{-1}(x, g) := (x, \phi_x(e)^{-1}g).$$

Since every G -principal bundle is locally trivial this shows that we can always find local inverses in the general case and since inverses are unique they fit together to define a global inverse. $\square$

To any fiber bundle with structure group G one can canonically associate a G -principal bundle, obtained by applying a general construction, which can be used to change the fiber of a given bundle. We now recall this construction of changing the fiber. Let $p: E \rightarrow B$ be a fiber bundle with fiber F and structure group G and let F' be another space upon which G acts effectively. Then we can construct a new bundle as follows. Choose an open cover $(U_\alpha)_{\alpha \in A}$ of B together with local trivialisations $\phi_\alpha: U_\alpha \times F \rightarrow p^{-1}(U_\alpha)$. Let $\tau_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow G$ be the corresponding transition functions defined by

$$(\phi_\beta^{-1} \circ \phi_\alpha)(x, f) = (x, \tau_{\alpha\beta}(x) \cdot f).$$

Define

$$E' := \left(\coprod_{\alpha \in A} U_\alpha \times F' \right) / \sim$$

where the equivalence relation $\sim$ is defined by

$$(b, f', \beta) \sim (b, \tau_{\alpha, \beta}(b) \cdot f', \alpha)$$

and let $p' : E' \rightarrow B$, $[(b, f', \alpha)] \mapsto b$. Then the maps

$$\psi_\alpha : U_\alpha \times F' \rightarrow (p')^{-1}(U_\alpha), (b, f') \mapsto [(b, f', \alpha)].$$

define local trivialisations of p' with the same transition functions $\tau_{\alpha\beta}$, giving $p' : E' \rightarrow B$ the structure of a fiber bundle with fiber F' and structure group G . We say that this new bundle is obtained from $E \rightarrow B$ by *changing the fiber* from F to F' (with the G -action on F' being implicit). Note that we can also obtain $E \rightarrow B$ from $E' \rightarrow B$ by changing the fiber from F' back to F . Thus no information is lost in this process. Considering the special case $F' = G$ with G acting on itself by left translations, we call the bundle obtained from $E \rightarrow B$ by changing the fiber from F to G the *associated principal bundle*. Morally speaking if we want to understand the bundle $E \rightarrow B$ it is sufficient to understand its associated principal bundle, as it contains the same information as we have seen above. The benefit of studying principal bundles is that they are highly symmetric objects, and thus usually easier to understand. For example many constructions simplify for principal bundles, such as the concept of universal bundles as we will see in the next subsection. Another such instance is the construction for changing the fiber from above, which can be carried out more directly in the case of principal bundles and has the benefit of also allowing for non-effective group actions. This is known as the *Borel Construction*.

Lemma 1.4. *Let $p : P \rightarrow B$ be a G -principal bundle and F a space upon which G acts from the left. Then*

$$E := (P \times F) / \sim, \quad (y, f) \sim (y \cdot g, g^{-1}f)$$

together with the map $\pi : E \rightarrow B$ given by $[(y, f)] \mapsto p(y)$ is a fiber bundle with fiber F and structure group G . Moreover, if the action of G on F is effective, this bundle is isomorphic to the bundle obtained from $P \rightarrow B$ by changing the fiber from G to F using the construction outlined above.

Proof. To see that $P \times_G F$ is locally trivial let $\psi : U \times G \rightarrow P$ be a local trivialisation. Then the homeomorphism

$$\psi \times \text{id}_F : U \times G \times F \rightarrow p^{-1}(U) \times F$$

descends to a homeomorphism

$$\tilde{\kappa} : (U \times G \times F) / \sim \rightarrow (p^{-1}(U) \times F) / \sim = \pi^{-1}(U),$$

where the equivalence relation on $U \times G \times F$ is given by $(x, g, f) \sim (x, gh, h^{-1}f)$. Moreover the map

$$U \times G \times F \rightarrow U \times F, (x, g, f) \mapsto (x, gf)$$

descends to a homeomorphism $(U \times G \times F) / \sim \rightarrow U \times F$ with inverse given by $(x, f) \mapsto [(x, e, f)]$.

Under this homeomorphism $\tilde{\kappa}$ corresponds to a homeomorphism $\kappa : U \times F \rightarrow \pi^{-1}(U)$, which is explicitly given by $(x, f) \mapsto [(\psi(x, e), f)]$ and

$$\begin{array}{ccc} U \times F & \xrightarrow{\kappa} & \pi^{-1}(U) \\ & \searrow \text{proj.} & \downarrow \pi \\ & & U \end{array}$$

commutes. So $P \times_G F$ is locally trivial. To see that this construction agrees with the change of fiber construction from above if G acts effectively we have to compute the transition functions of the local trivialisations. Let $U_\alpha, U_\beta \subseteq B$ be two trivialising neighbourhoods for P with local trivialisations ψ_α, ψ_β and transition function $\tau_{\alpha, \beta} : U_\alpha \cap U_\beta \rightarrow G$. Moreover let $\kappa_\alpha, \kappa_\beta$ be the corresponding local trivialisations for $P \times_G F$ as constructed above. Then we compute

$$(\kappa_\alpha^{-1} \circ \kappa_\beta)(x, f) = \kappa_\alpha^{-1}([\psi_\beta(x, e), f]) = (x, \tau_{\alpha, \beta}(x)f).$$

Thus the statement follows from the fact that fiber bundles with structure group are uniquely determined by the transition functions of local trivialisations. $\square$

In view of the above construction one usually denotes the bundle obtained from $P \rightarrow B$ by changing the fiber from G to F as $P \times_G F \rightarrow B$. The Borel construction is natural in the following sense.

Lemma 1.5. *Let $p: P \rightarrow B$ be a G -principal bundle, F a space upon which G acts from the left and $f: B' \rightarrow B$ a continuous map. Then*

$$(f^*P) \times_G F \cong f^*(P \times_G F).$$

Proof. An explicit model for the pullback bundle is

$$f^*P := \{(y, b') \in P \times B' \mid p(y) = f(b')\}.$$

Then

$$(f^*P) \times_G F = \{(y, b', f) \in P \times B' \times F \mid p(y) = f(b')\} / \sim,$$

for $(y, b', f) \sim (y \cdot g, b', g^{-1} \cdot f)$. Moreover

$$\begin{aligned} f^*(P \times_G F) &= \{([y, f], b') \in P \times_G F \times B' \mid p(y) = f(b')\} \\ &\cong \{(y, f, b') \in P \times F \times B' \mid p(y) = f(b')\} / \sim \end{aligned}$$

for $(y, f, b') \sim (y \cdot g, g^{-1} \cdot f, b')$. Clearly these are isomorphic. $\square$

Lets look at some concrete examples for the theory developed above.

Example 1.6. For simplicity we assume all base spaces B to be smooth manifolds in this example.

- i) Let $E \rightarrow B$ be a real rank n vector bundle. Choosing a bundle metric gives $E \rightarrow B$ the structure of a fiber bundle with fiber $\mathbb{R}^n$ and structure group $O(n)$. The associated principal bundle is explicitly given by the *orthonormal frame bundle* $O(E)$ of E , whose fiber over a point $x \in B$ is the space of all ordered orthonormal bases of E_x . Local trivialisations of the canonical map $p: O(E) \rightarrow B$ can be defined as follows. Choose an orthonormal local frame $e_1, \dots, e_n$ of E defined over an open set $U \subseteq B$. Then a trivialisation $\phi: U \times O(n) \rightarrow p^{-1}(U)$ is given by

$$(x, A) \mapsto \left(\sum_{i=1}^n a_{i1} e_i(x), \dots, \sum_{i=1}^n a_{in} e_i(x) \right),$$

where $A = (a_{ij})$, i.e. $\phi(x, A)$ is the basis of $E_x = p^{-1}(x)$, whose coordinates in the frame (e_i) are precisely the columns of the matrix A . The transition functions of these trivialisations agree with the transition functions of the trivialisations of E . Again, since the transition functions uniquely determine the bundle, we conclude that $O(E)$ is in fact isomorphic to the associated principle bundle as defined at the beginning of this section.

- ii) An orientation of a real vector bundle allows us to further reduce the structure group to $SO(n)$. Similiarly, the associated principal bundle is then given by the *oriented orthonormal frame bundle* $SO(E)$ of E , whose fiber over a point $x \in B$ is the space of all oriented ordered orthonormal bases of E_x .
- iii) An example, which will be of relevance later is the following. Let $P \rightarrow B$ be an S^1 -principal bundle. We can change the fiber of P from S^1 to D^2 to obtain another bundle $E \rightarrow B$, where S^1 acts on D^2 by rotations. Since this action just extends the action of $S^1 = \partial D^2$ on itself we see that $P \cong \partial E$. So in particular the total space of any S^1 -principal bundle is null-bordant.

1.2 Universal Bundles and Classifying Spaces

A natural question is how many bundles exist... again easier for principal bundles... For simplicity we assume all base spaces to be CW-complexes in this section.

Definition 1.7. A G -principal bundle $E \rightarrow B$ is called universal if E is weakly contractible as a topological space.

The significance of this definition is established by the following theorem. We say that two morphisms of G -principal bundles are G -homotopic if they are homotopic through morphisms of G -principal bundles.

Theorem 1.8. *Let $E \rightarrow B$ be a universal G -principal bundle. Then for any other G -principal bundle $P \rightarrow X$ there exists a morphism $P \rightarrow E$ of G -principal bundles, which is unique up to G -homotopy.*

Before showing the Theorem let us discuss its implications. Firstly we can equivalently formulate the Theorem by saying that $E \rightarrow B$ is a terminal object in the homotopy category of G -principal bundles over CW-complexes and thus in particular uniquely determined (up to G -homotopy equivalence). This justifies writing $EG := E$ and $BG := B$. Since any bundle morphism $(F, f): P \rightarrow EG$ induces a morphism $P \rightarrow f^*EG$ (and vice versa, which is true for arbitrary fiber bundles) we moreover conclude from Proposition 1.3 that

$$P \cong f^*EG.$$

We call f a **classifying map** for P . By homotopy invariance of the pullback we get a well-defined bijection

$$\phi_X: [X, BG] \rightarrow \mathcal{P}_G(X), [f] \mapsto [f^*EG],$$

where $\mathcal{P}_G(X)$ denotes the set of isomorphism classes of G -principal bundles over X . To see that ϕ_X is indeed bijective note that the existence result in Theorem ?? shows that ϕ_X is surjective and the uniqueness result shows that it is injective. Moreover ϕ_X is clearly natural in X . Thus we have shown.

Corollary 1.9. *The maps ϕ_X define a natural isomorphism of contravariant functors*

$$[-, BG], \mathcal{P}_G(-): \mathbf{HCW} \rightarrow \mathbf{Set},$$

where $\mathbf{HCW}$ denotes the homotopy category of CW-complexes.

Theorem 1.8 will be an application of the following Theorem.

Theorem 1.10. *Let $P \rightarrow X$ be a G -principal bundle and $A \subseteq X$ a subcomplex. Then any morphism $P|_A \rightarrow EG$ can be extended to a morphism $P \rightarrow EG$.*

A proof of this Theorem based on an inductive cell by cell approach can be found in [Ebe12, Theorem 2.7.1].

Proof of Theorem 1.8. Applying Theorem 1.10 to $A = \emptyset$ shows the existence of a morphism $P \rightarrow EG$. Now suppose we are given morphisms $F_0, F_1: P \rightarrow EG$. Consider the G -principal bundle

$$\pi \times \text{id}: P \times [0, 1] \rightarrow X \times [0, 1],$$

where $\pi: P \rightarrow X$ is the bundle projection (this is just the pullback bundle of P under the projection $X \times [0, 1] \rightarrow X$). Then F_0 and F_1 define a morphism of G -principal bundles

$$(F_0, F_1): P \times \{0, 1\} = (P \times [0, 1])|_{X \times \{0, 1\}} \rightarrow EG,$$

which by Theorem 1.10 can be extended to $P \times [0, 1]$, thus yielding a G -homotopy between F_0 and F_1 . $\square$

of course for this theory to be of any use we need such universal bundles to exist...

Theorem 1.11 (Milnor construction). *For any topological group G there exists a universal G -principal bundle $EG \rightarrow BG$, where BG is a CW-complex.*

understood principal bundles, how can we recover information about general bundles by changing the fiber

Corollary 1.12. *Let G be a topological group acting effectively on a space F . Then for any CW-complex X the map*

$$[X, BG] \rightarrow \mathcal{F}_{F,G}(X), [f] \mapsto [f^*(EG \times_G F)]$$

is a natural bijection, where $\mathcal{F}_{F,G}(X)$ denotes the set of isomorphism classes of fiber bundles with fiber F and structure group G over X .

Proof. By 1.9 the map

$$[X, BG] \rightarrow \mathcal{P}_G(X), \quad [f] \mapsto [f^* EG]$$

is a natural bijection and by Lemma 1.5

$$\mathcal{P}_G(X) \rightarrow \mathcal{F}_{F,G}(X), \quad [P] \mapsto [P \times_G F]$$

is a natural bijection. Composing both thus yields the natural bijection

$$[X, BG] \rightarrow \mathcal{F}_{F,G}(X), \quad [f] \mapsto [(f^* EG) \times_G F].$$

So the statement follows by again applying Lemma 1.5. $\square$

Let us look at some examples...

Example 1.13.

- (i) homotopy groups of BG ($\pi_k(BG) \cong \pi_{k-1}(G)$ for $k \geq 1$ and $\pi_0(BG) = *$, since $BG = \pi(EG)$, where $\pi: EG \rightarrow BG$ projection)
- (ii) $BO(n)$ and $EO(n)$ explicit via Stiefel Manifolds (compare n lab) + $EO(n) \times_{O(n)} \mathbb{R}^n \cong \gamma_n$ for γ_n tautological bundle over grassmanian + orientable vector bundles classified by $BSO(n)$ (universal vector bundle over this denoted by γ_n^{SO})
- (iii) Analogous for complex case??
- (iv) Special cases $\mathbb{R}P^\infty$ and $\mathbb{C}P^\infty$
- (v) $M \rightarrow B\pi_1(M)$ induces iso on π_1 (brauch ich gar nicht...)

1.3 Stiefel-Whitney and Chern-Classes

We have seen how Another fundamental question is of course how to decide, whether two bundles are isomorphic. As usual in topology, one tries to define certain invariants posing as an obstruction... For vector bundles (or more generally principal bundles), ... characteristic classes. The general definition of a characteristic class is as follows. Let $\mathbb{K} = \mathbb{R}, \mathbb{C}$ and write $\text{Vect}_{\mathbb{K}}(B)$ for the set of isomorphism classes of $\mathbb{K}$ -vector bundles over a space X . Then $\text{Vect}_{\mathbb{K}}(-)$ defines a contravariant functor $\mathbf{Top} \rightarrow \mathbf{Set}$. A characteristic class c for $\mathbb{K}$ -vector bundles with values in the cohomology theory H^* is a natural transformation from the functor $\text{Vect}_{\mathbb{K}}(-)$ to the functor H^* regarded as a functor with values in $\mathbf{Set}$. So in more concrete terms c assigns to each $\mathbb{K}$ -vector bundle $E \rightarrow B$ a cohomology class $c(E) \in H^*(B)$ depending only on the isomorphism class of E , such that for any continuous map $f: B' \rightarrow B$ the following naturality condition holds

$$c(f^* E) = f^* c(E).$$

So in particular, using the theory of classifying spaces developed above, we see that for a real (resp. complex) rank k vector bundle $E \rightarrow B$, the class $c(E)$ is uniquely determined by the value $c(\xi_k^\infty) \in H^*(BO(k))$ for the universal bundle ξ_k^∞ introduced in example...

For real vector bundles the most important characteristic classes are the so called *Stiefel-Whitney classes*, which can be completely characterized by a few axioms.

Definition 1.14 (Stiefel-Whitney Classes, HATCHER). There is a unique sequence of characteristic classes $w_1, w_2, \dots$ for real vector bundles with values in $H^*(-; \mathbb{Z}_2)$, such that

- i) $w_i(E) \in H^i(B; \mathbb{Z}_2)$ for any real bundle $E \rightarrow B$,
- ii) $w_i(E) = 0$ for $i > \text{rank}(E)$,
- iii) $w(E_1 \oplus E_2) = w(E_1) \smile w(E_2)$, where $w = 1 + w_1 + w_2 + \dots$,
- iv) For the canonical line bundle $\gamma_1 \rightarrow \mathbb{R}P^\infty$, $w_1(\gamma_1)$ is the generator of $H^1(\mathbb{R}P^\infty; \mathbb{Z}_2) \cong \mathbb{Z}_2$.

Note that by property ii) the sum $w = 1 + w_1 + w_2 + \dots$ is always finite. This class $w(E)$ is called the *total Stiefel-Whitney class* of E . For obvious reasons one sometimes writes $w_0(E) := 1 \in H^0(B; \mathbb{Z}_2)$. The analogues for complex vector bundles are known as *Chern classes*.

Definition 1.15 (Chern Classes, HATCHER). There is a unique sequence of characteristic classes $c_1, c_2, \dots$ for complex vector bundles with values in $H^*(-; \mathbb{Z})$, such that

- i) $c_i(E) \in H^{2i}(B; \mathbb{Z})$ for any complex bundle $E \rightarrow B$,
- ii) $c_i(E) = 0$ for $i > \text{rank}(E)$,
- iii) $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$, where $c = 1 + c_1 + c_2 + \dots$,
- iv) For the canonical line bundle $\tau_1 \rightarrow \mathbb{C}P^\infty$, $c_1(\tau_1)$ is a generator of $H^2(\mathbb{C}P^\infty; \mathbb{Z}) \cong \mathbb{Z}$ specified in advance.

Again $c = 1 + c_1 + c_2 + \dots$ is well-defined and is called the *total Chern class*. Again one sometimes writes $c_0(E) := 1 \in H^0(B; \mathbb{Z})$. See [HATCHER] for a proof of existence and uniqueness.

Example 1.16.

- i) Let $\iota: \mathbb{R}P^n \rightarrow \mathbb{R}P^\infty$ be the canonical inclusion. Then the pullback $\iota^*\gamma_1 \cong \gamma_1^n$ is the tautological line bundle $\gamma_1^n \rightarrow \mathbb{R}P^n$. Since ι induces an isomorphism on cohomology we conclude from naturality that $w_1(\gamma_1^n) = \iota^*w_1(\gamma_1)$ is the generator of $H^1(\mathbb{R}P^n; \mathbb{Z}_2) \cong \mathbb{Z}_2$. Similarly $c_1(\tau_1^n)$ of the tautological complex line bundle $\tau_1^n \rightarrow \mathbb{C}P^n$ is a generator of $H^2(\mathbb{C}P^n; \mathbb{Z}) \cong \mathbb{Z}$.
- ii) Let $\varepsilon \rightarrow B$ be a trivial real bundle over some space B . Then $\varepsilon \cong f^*\varepsilon$ for a constant map $f: B \rightarrow B$ and thus $w_i(\varepsilon_n) = f^*w_i(\varepsilon) = 0$ for all $i \geq 1$. Similarly all Chern classes for complex trivial bundles vanish.
- iii) Let $TS^n \rightarrow S^n$ be the tangent bundle of the sphere and let $\nu_n \rightarrow S^n$ be the normal bundle. Since ν_n and $TS^n \oplus \nu_n$ are trivial we conclude from the example above and property iii) in Definition 1.14 that

$$1 = w(TS^n \oplus \nu_n) = w(TS^n) \smile w(\nu_n) = w(TS^n) \smile 1 = w(TS^n).$$

Thus $w_i(TS^n) = 0$ for all $i \geq 1$.

In the last example we have seen that although all Stiefel-Whitney classes of a bundle vanish, the bundle need not be trivial. However for one-dimensional real (resp. complex) bundles it turns out that they are already completely classified by their first Stiefel-Whitney class (resp. Chern class):

Theorem 1.17 (Hatcher, Proposition 3.10). *Let X be a CW-complex. Then the maps*

$$w_1: \text{Vect}_{\mathbb{R}}(X) \rightarrow H^1(X; \mathbb{Z}_2)$$

and

$$c_1: \text{Vect}_{\mathbb{C}}(X) \rightarrow H^2(X; \mathbb{Z})$$

are bijections.

This allows us to define line bundles by only prescribing their first Stiefel-Whitney class (resp. Chern class), which is often technically easier than directly writing down this bundle. While the first Stiefel-Whitney class of higher dimensional bundles does not identify their isomorphism type it does characterize another important invariant, namely orientability:

Proposition 1.18 (Hatcher, Prop 3.11). *Let X be a CW-complex. A real vector bundle $E \rightarrow X$ is orientable if and only if $w_1(E) = 0$.*

Similarly vanishing of the second Stiefel Whitney class $w_2(E)$ corresponds to the existence of another structure on E , namely a spin structure.

Definition 1.19. Let $n \geq 3$.

- (i) The *spin group* $\text{Spin}(n)$ is defined as the universal cover $\rho: \text{Spin}(n) \rightarrow \text{SO}(n)$.
- (ii) Let $E \rightarrow B$ be an oriented real vector bundle equipped with a fiber metric and let $P_{\text{SO}(n)}$ be its oriented orthonormal frame bundle. A *spin structure* on E is a $\text{Spin}(n)$ -principal bundle $P_{\text{Spin}(n)}$ together with a map

$$\phi: P_{\text{Spin}(n)} \rightarrow P_{\text{SO}(n)}$$

of principal bundles, which is ρ -equivariant (see Definition 1.2).

- (iii) An orientable manifold M is called *spin* if its tangent bundle TM admits a spin structure and is called *totally non-spin* if the universal cover of M is not spin.

Note that the existence of a spin structure does not depend on the chosen orientation or fiber metric, since different choices yield isomorphic oriented orthonormal frame bundles. In fact the only obstruction to the existence of a spin structure is the second Stiefel-Whitney class, as mentioned above.

Theorem 1.20 (Lawson, Micheaslon). *Let X be a CW-complex. An orientable real vector bundle $E \rightarrow X$ admits a spin structure if and only if $w_2(E) = 0$.*

Recall that a manifold M is orientable if and only if its tangent bundle TM restricted to any embedded loop $S^1 \subseteq M$ is trivial. Or in other words “ TM does not contain any Moebius Bands”. A similar characterization can be given for spin manifolds, which is why the property of being spin is sometimes referred to as a higher dimensional analogue of orientability.

Proposition 1.21 (Georg Paper, Proposition 2.6). *Let M be a connected oriented manifold of dimension at least 5.*

- i) *M is spin if and only if for every embedded surface $S \subseteq M$ the restriction $TM|_S$ of the tangent bundle to S is trivial.*
- ii) *M is totally non-spin if and only if there exists an embedded sphere $S^2 \subseteq M$, such that the restriction $TM|_{S^2}$ of the tangent bundle to this sphere is trivial.*

Proof. Since we will not be making use of the first characterization of spin manifolds, we shall only proof part ii) of the proposition and refer the reader to [Georg Paper] for a proof of part i). ... (Proof) $\square$

Remark 1.22. Let $E \rightarrow X$ be a real vector bundle over a CW-complex. If an orientation is given, E can be classified by a map $X \rightarrow BSO(n)$ (see example 1.13). Similarly, if a spin structure exists, E can be classified by a map $X \rightarrow BSpin(n)$, which can be seen as follows. Let $\phi: P_{Spin(n)} \rightarrow P_{SO(n)}$ be a spin structure on E and let $f: X \rightarrow BSpin(n)$ be the classifying map for $P_{Spin(n)}$. Define a vector bundle

$$\gamma_n^{Spin} := ESpin(n) \times_{Spin(n)} \mathbb{R}^n$$

using Lemma 1.4, where $Spin(n)$ acts on $\mathbb{R}^n$ by virtue of the covering map $\rho: Spin(n) \rightarrow SO(n)$ and the canonical action of $SO(n)$. From Lemma 1.5 we obtain

$$f^* \gamma_n^{Spin} \cong P_{Spin(n)} \times_{Spin(n)} \mathbb{R}^n.$$

Moreover, the spin structure ϕ induces an isomorphism $P_{Spin(n)} \times_{Spin(n)} \mathbb{R}^n \rightarrow P_{SO(n)} \times_{SO(n)} \mathbb{R}^n \cong E$ of vector bundles by

$$[(x, v)] \mapsto [(\phi(x), v)].$$

This map is clearly a linear bijection on fibers, hence an isomorphism of vector bundles.

Finally we record another scenario in which the vanishing of all Stiefel-Whitney classes of a bundle is sufficient for triviality.

Lemma 1.23. *Let $E \rightarrow S$ be a real vector bundle of rank at least 3 over a two dimensional CW-complex S . Then E is trivial if and only if $w_1(E) = w_2(E) = 0$.*

Proof. Combining Theorem 1.20 and Remark 1.22 we conclude that there exists a classifying map $f: S \rightarrow BSpin(n)$ for E , where n is the rank of E . From example 1.13 it follows that $\pi_k(BSpin(n)) = 0$ for $k \geq 2$. Since S is two dimensional the map f is homotopic to a constant ([Hatcher, Lemma 4.6]), therefore $E \cong f^* \xi_{Spin}$ is trivial. $\square$

2 Positive Scalar Curvature

2.1 Overview

brief definition, maybe intuition and obstructions/background etc. based on overview paper

psc is somehow sweetspot for rigidity and flexibility

Probably the most important existence result for positive scalar curvature metrics is the following theorem, which has been shown independently by Schoen and Yau [SY79] as well as Gromov and Lawson [GL80]. We refer the reader to the Appendix for a brief introduction of surgery and cobordism theory.

Theorem 2.1 (Gromov, Lawson, 1980, Chernysh Theorem Georg Paper). *Let M be a compact manifold, which admits a psc metric. Then any manifold M' obtained from M by surgery of codimension at least 3 also admits a psc metric.*

We remark that the construction of the psc metric on M' only changes the given psc metric on M locally near the embedding along which the surgery is performed. In order to apply this theorem we need a way to control the dimension of the surgeries performed or equivalently the indices of handles in a given cobordism. This question is completely independent of scalar curvature and has for example been studied quite thoroughly in the context of the h -cobordism theorem. One central tool to control the indices of the handles in a cobordism is the following theorem due to Wall

Theorem 2.2 (Wall, 1971). (*Wall Theorem*)

Combining the above theorem with the Surgery Theorem 2.1, we can make the following observation. If M_0 admits a psc metric and the inclusion $M_1 \rightarrow W$ is 2-connected, then M_1 also admits a psc metric. In fact, using the conclusion of Theorem 2.2 it is possible to obtain an even stronger result. We say that a metric g on a manifold M with boundary is of *product form near the boundary*, if there is some collar neighbourhood $\iota: \partial M \times [0, 1) \rightarrow M$, such that $\iota^*g = g|_{\partial M} + dt^2$, where $g|_{\partial M}$ is the induced metric on ∂M .

Theorem 2.3 (extending psc to cobordism, Georg, Gaja). *Let W^{n+1} be a cobordism from M_0 to M_1 , such that the inclusion $M_1 \rightarrow W$ is 2-connected. Then any psc metric on M_0 can be extended to a psc metric on W , which is of product form near the boundary.*

2.2 Existence Results for psc metrics

In view of the above discussion, one strategy to construct a psc metric on a given closed manifold M_1 of dimension at least 5 is to first find a cobordism W from some psc manifold M_0 to M_1 and then modify this cobordism by surgeries to obtain another cobordism W' from M_0 to M_1 , such that the inclusion $M_1 \rightarrow W'$ is 2-connected. Of course, such a modification by surgeries may not always be possible, so our next goal is to explain certain scenarios in which it is, one in the spin case and one in the totally non-spin case (see Definition ??).

Theorem 2.4. *Let W^{n+1} be a cobordism from M_0 to M_1 , $n \geq 5$, such that W is spin. Suppose that M_1 is connected, the inclusion $M_1 \rightarrow W$ induces an isomorphism on π_1 and that M_0 carries a psc-metric. Then M_1 also carries a psc-metric. In fact the psc-metric on M_0 can be extended to a psc-metric on W , which is of product form near the boundary.*

Theorem 2.5. *Let W^{n+1} be an oriented cobordism from M_0 to M_1 , $n \geq 5$. Suppose that M_1 is a connected totally non-spin manifold, the inclusion $M_1 \rightarrow W$ induces an isomorphism on π_1 and that M_0 carries a psc-metric. Then M_1 also carries a psc-metric. In fact the psc-metric on M_0 can be extended to a psc-metric on W , which is of product form near the boundary.*

We first show the totally non-spin case.

Proof of Theorem 2.5. Without loss of generality assume W to be connected as well (otherwise only consider the connected component containing M). We first show that $\pi_2(W, M)$ is finitely generated. Let $\tilde{M}$ and $\tilde{W}$ be the universal covers of M and W respectively. Then the inclusion $M \rightarrow W$ lifts to a map $\tilde{M} \rightarrow \tilde{W}$ by virtue of which we can consider the relative homotopy group $\pi_2(\tilde{W}, \tilde{M})$. It follows from the Hurewicz Theorem that

$$\pi_2(\tilde{W}, \tilde{M}) \cong H_2(\tilde{W}, \tilde{M}).$$

By ??? $H_2(\tilde{W}, \tilde{M})$ is finitely generated as a $\mathbb{Z}[\pi_1(M)]$ module (??) hence the same holds for $\pi_2(\tilde{W}, \tilde{M})$. Now consider the following diagram where the rows are exact.

$$\begin{array}{ccccc} \pi_2(\tilde{M}) & \longrightarrow & \pi_2(\tilde{W}) & \twoheadrightarrow & \pi_2(\tilde{W}, \tilde{M}) \\ \downarrow \cong & & \downarrow \cong & & \downarrow \\ \pi_2(M) & \longrightarrow & \pi_2(W) & \twoheadrightarrow & \pi_2(W, M) \end{array}$$

The two rightmost horizontal maps are surjective by exactness, since $\pi_1(\tilde{M}) = 0$ and $\pi_1(M) \rightarrow \pi_1(W)$ is an isomorphism. A diagram chase now shows that $\pi_2(\tilde{W}, \tilde{M}) \rightarrow \pi_2(W, M)$ is an isomorphism.

Next find lifts with... Let

$$F: W \rightarrow B := BSO(n+1)$$

be a classifying map for the tangent bundle TW and set $f := F \circ i$, where $i: M_1 \rightarrow W$ is the inclusion. Then f is a classifying map for $TM_1 \oplus \mathbb{R}$ with $\mathbb{R}$ the trivial line bundle. We claim that f is surjective on π_2 . Since $\pi_2(B) \cong \pi_1(SO(n+1)) \cong \mathbb{Z}_2$ it suffices to show that $\pi_2(f)$ is non-zero. By Proposition 1.21 there exists an embedding $\iota: S^2 \rightarrow M_1$, such that $w_2(\iota^*TM_1) \neq 0$. We compute

$$w_2((f \circ \iota)^*\gamma_n^{\text{SO}}) = w_2(\iota^*(TM_1 \oplus \mathbb{R})) = w_2(\iota^*TM_1) \neq 0.$$

Therefore $f \circ \iota$ represents a non-trivial element in $\pi_2(B)$ showing the claim. Now consider the following diagram, where the row is exact.

$$\begin{array}{ccccc} \pi_2(M_1) & \xrightarrow{i_*} & \pi_2(W) & \longrightarrow & \pi_2(W, M_1) \\ & \searrow f_* & \downarrow F_* & & \\ & & \pi_2(B) & & \end{array}$$

Using surjectivity of f_* , a diagram chase shows that we can find lifts $\alpha_1, \dots, \alpha_l \in \pi_2(W)$ of the generators $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}} \in \pi_2(W, M_1)$, which lie in the kernel of F_* . Moreover, since $n \geq 5$ the classes α_i can be represented by embeddings

$$\phi_i: S^2 \rightarrow W.$$

We show that the normal bundle ν_i of ϕ_i is trivial. Since W is oriented it follows from Proposition 1.18 that $w_1(\phi_i^*TW) = 0$. Furthermore, $F \circ \phi_i$ is null-homotopic by construction, such that

$$w_2(\phi_i^*TW) = w_2((F \circ \phi_i)^*\gamma_n^{\text{SO}}) = 0.$$

Using $w_k(TS^2) = 0$ (see Example 1.16 iii)) we compute

$$w_k(\nu_i) = w_k(\nu_i \oplus TS^2) = w_k(\phi_i^*TW) = 0$$

for $k = 1, 2$. Thus Lemma 1.23 shows that ν_i is indeed trivial. This allows us to perform surgery to obtain a new cobordism W' from M_0 to M_1 with

$$\pi_2(W', M_1) = 0$$

and $\pi_1(W', M_1) \cong \pi_1(W, M_1) = 0$, see Proposition A.2 (ii). In other words, the inclusion $M_1 \rightarrow W'$ is 2-connected. So by Theorem 2.3 there exists a psc metric g on W' , which is of product form near the boundary. To construct a psc metric on the original cobordism W , observe that since W' was obtained from W by surgeries of dimension 2, W can be obtained from W' by surgeries of codimension 3. Therefore the surgery theorem 2.1 applies to construct a psc metric on W . Since this construction only changes the given metric g locally near the surgery embeddings, the resulting metric on W is still of product form near the boundary. $\square$

Proof of Theorem 2.4. Again $\pi_2(W, M)$ is finitely generated as a $\mathbb{Z}[\pi_1(M)]$ -module. Let $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ be such generators. We could argue as in the proof of Theorem 2.5, replacing $BSO(n+1)$ by $B\text{Spin}(n+1)$ and observing that the corresponding f is trivially surjective on π_2 as $\pi_2(B\text{Spin}(n+1)) \cong \pi_1(\text{Spin}(n+1)) = 0$. However this just amounts to the following more direct argument. Choose lifts $\alpha_1, \dots, \alpha_l \in \pi_2(W)$ of $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ and represent α_i by an embedding $\phi_i: S^2 \rightarrow W$. Then

$$w_k(\phi_i^*TW) = \phi_i^*w_k(TW) = 0$$

for $k = 1, 2$ since W is spin. The rest of the proof is now as in Theorem 2.5. $\square$

Remark 2.6. The proofs of Theorem 2.4 and 2.5 are very similar and in fact both statements are a consequence of a more general result. This unifying approach can be formulated using the language of tangential 2-types, which we shall not discuss further in this thesis. See for example [Georg Paper etc.].

In the main construction in Chapter 4 we will invoke Theorem 2.5 to construct a psc metric, so let us conclude this section by discussing some basic results for identifying totally non-spin manifolds.

Lemma 2.7. (i) *The product of a totally non-spin manifold with any non-empty manifold is totally non-spin.*

(ii) *A manifold containing a totally non-spin submanifold of codimension 0 is totally non-spin.*

(iii) *If M is totally non-spin and of dimension $n \geq 3$, then $M - D^n$ is totally non-spin for any embedded disk $D^n \subseteq \text{int}(M)$ in the interior of M .*

Remark 2.8. All statements of the above lemma remain valid if “totally non-spin” is replaced by “non-spin”. The corresponding proofs are just simpler versions of the arguments given below.

Proof. For (i) let M_1 and M_2 be manifolds with M_1 totally non-spin. Then the universal cover of the product $M_1 \times M_2$ is given by the product $\tilde{M}_1 \times \tilde{M}_2$ of the respective universal covers. Denote by

$$\pi_1: \tilde{M}_1 \times \tilde{M}_2 \rightarrow \tilde{M}_1, \quad \pi_2: \tilde{M}_1 \times \tilde{M}_2 \rightarrow \tilde{M}_2$$

the projections and let $\iota: \tilde{M}_1 \rightarrow \tilde{M}_1 \times \tilde{M}_2$ be an inclusion for some choice of basepoint in $\tilde{M}_2$. We compute

$$\begin{aligned} \iota^* w_2(T(\tilde{M}_1 \times \tilde{M}_2)) &= \iota^* w_2(\pi_1^* T\tilde{M}_1 \oplus \pi_2^* T\tilde{M}_2) \\ &= w_2((\pi_1 \circ \iota)^* T\tilde{M}_1 \oplus (\pi_2 \circ \iota)^* T\tilde{M}_2) \\ &= w_2(T\tilde{M}_1) \neq 0, \end{aligned}$$

where we used that $\pi_1 \circ \iota$ is the identity and that $\pi_2 \circ \iota$ is constant, such that $(\pi_2 \circ \iota)^* T\tilde{M}_2$ is trivial. Thus $w_2(T(\tilde{M}_1 \times \tilde{M}_2)) \neq 0$.

For (ii) suppose that $N \subseteq M$ is some totally non-spin codimension 0 submanifold of the manifold M . Let $\iota: N \rightarrow M$ be the inclusion and consider its lift to the universal covers:

$$\begin{array}{ccc} \tilde{N} & \xrightarrow{\tilde{\iota}} & \tilde{M} \\ \downarrow & & \downarrow \\ N & \xrightarrow{\iota} & M \end{array}$$

Since $\tilde{\iota}$ is a local diffeomorphism, the differential $d\tilde{\iota}: T\tilde{N} \rightarrow T\tilde{M}$ is a fiber-wise isomorphism, such that the following is a pullback diagram

$$\begin{array}{ccc} T\tilde{N} & \xrightarrow{d\tilde{\iota}} & T\tilde{M} \\ \downarrow & & \downarrow \\ \tilde{N} & \xrightarrow{\tilde{\iota}} & \tilde{M} \end{array}$$

Therefore $\tilde{\iota}^* w_2(T\tilde{M}) = w_2(T\tilde{N}) \neq 0$ and so $w_2(T\tilde{M}) \neq 0$.

For (iii) assume without loss generality that M is connected. Denote by $B^n := \text{int}(D^n)$ the open ball. Then $M - D^n$ is the interior of $M - B^n$ from which we conclude that $M - D^n$ is totally non-spin if and only if $M - B^n$ is totally non-spin. This follows using an analogous argument as in (ii) by noting that for a manifold X with boundary the universal cover of $\text{int}(X)$ is given by $\text{int}(\tilde{X})$ and the inclusion $\text{int}(\tilde{X}) \rightarrow \tilde{X}$ is a homotopy equivalence. We will now show that $M - B^n$ is totally non-spin. Let $p: \tilde{M} \rightarrow M$ be the universal cover of M and write $M_0 := M - B^n$. We claim that the universal cover of M_0 agrees with the covering

$$\hat{M}_0 = \tilde{M}|_{M_0} = \tilde{M} - p^{-1}(B^n) \xrightarrow{p} M_0.$$

First of all $\hat{M}_0$ is connected as $n \geq 2$. We now have to show that $\hat{M}_0$ is simply connected. Let $F = p^{-1}(x)$ be some fiber of p . From the classification of coverings we see that

$$p^{-1}(B^n) \cong B^n \times F$$

since B^n is simply connected. If the fiber F is finite we can apply the Seifert-van-Kampen Theorem to conclude that $\tilde{M}$ with a single disk B^n removed is still simply connected, such that the statement follows inductively. If the fiber is not finite we can argue more generally as follows. Let $\phi: S^1 \rightarrow \hat{M}_0$ be some smooth loop. Certainly we can contract this loop in $\tilde{M}$, that is we can find some map $\Phi: D^2 \rightarrow \tilde{M}$ restricting to α on the boundary. We now show that Φ is homotopic rel S^1 to some map $\Phi_1: D^2 \rightarrow \tilde{M}$, such that the image of Φ_1 is already contained in $\hat{M}_0$, from which the claim follows. To construct such a homotopy let $0 \in B^n \subseteq M$ be the center of the embedded disk. After a preliminary homotopy we may assume that Φ is transversal to $p^{-1}(0)$. The crucial observation is that, since $n \geq 3$, this is equivalent to requiring that Φ does not intersect $p^{-1}(0)$. Now note that S^{n-1} is a strong deformation retract of $D^n - 0$ and applying such a deformation retraction on each disk $D^n - 0 \subseteq p^{-1}(D^n - 0)$ shows that $\hat{M}_0$ is a strong deformation retract of $\tilde{M} - p^{-1}(0)$. Let

$$r: \tilde{M} - p^{-1}(0) \rightarrow \hat{M}_0$$

be such a retraction and let

$$r_t: \tilde{M} - p^{-1}(0) \rightarrow \tilde{M} - p^{-1}(0)$$

be a homotopy rel S^1 from the identity to r (composed with the corresponding inclusion). Then $\Phi_t := r_t \circ \Phi$ defines a homotopy rel S^1 from Φ to some map Φ_1 whose image is contained in $\hat{M}_0$. This shows the claim and hence the universal cover of M_0 is given by

$$\hat{M}_0 = \tilde{M} - p^{-1}(B^n) \cong \tilde{M} - (B^n \times F).$$

Finally, the Mayer-Vietoris sequence of the pushout

$$\begin{array}{ccc} S^{n-1} \times F & \longrightarrow & D^n \times F \\ \downarrow & & \downarrow \\ \tilde{M} - (B^n \times F) & \xrightarrow{\iota} & \tilde{M} \end{array}$$

where all maps are inclusions, then shows that ι is injective on $H^2(-; \mathbb{Z}_2)$. Therefore

$$w_2(T\hat{M}_0) = w_2(\iota^*T\tilde{M}) = \iota^*w_2(T\tilde{M}) \neq 0.$$

□

A standard example of a non-spin manifold is $\mathbb{C}P^2$. Since $\mathbb{C}P^2$ is simply connected this is also an example of a totally non-spin manifold, which we will appeal to in section ...

Lemma 2.9. $\mathbb{C}P^2$ is non-spin.

(proof)

3 Enlargeability

In this section we discuss the concept of enlargeability, which is the central idea the results in this thesis are concerned with.

Definition 3.1. A closed oriented Riemannian n -manifold M is called enlargeable if for all $\varepsilon > 0$ there exists a Riemannian covering $\hat{M} \rightarrow M$ together with an ε -contracting (that is ε -Lipschitz) map $\hat{M} \rightarrow S^n$, which is constant outside a compact subset and of non-zero degree.

Note that this notion does not depend on the particular choice of Riemannian metric on M , since for closed manifolds all metrics are in bi-Lipschitz correspondence. Intuitively, the covering $\hat{M}$ can be pictured as an “expansion” of M itself (think for example of the usual sketches of coverings of S^1). Enlargeability now means that this expansion can be chosen arbitrarily large, such that it can be wrapped as tightly around a sphere as one wishes. Naively one might expect that for an enlargeable manifold M the universal cover $\tilde{M}$ - which in some sense is the “largest” covering - is hyperspherical, meaning that it admits an ε -contracting map $\tilde{M} \rightarrow S^n$, which is constant outside a compact subset and of non-zero degree for arbitrarily small $\varepsilon > 0$. However, Brunnbauer and Hanke have constructed counterexamples to this naive expectation in [BH, Theorem 1.4].

The simplest example of an enlargeable manifold is the Torus T^n , since it is covered by $\mathbb{R}^n$, which is clearly hyperspherical. More generally the same holds for any closed manifold admitting a metric of non-positive sectional curvature by the Cartan-Hadamard Theorem. Non-enlargeable manifolds are for example all simply connected manifolds.

The central fact about enlargeable manifolds is that they do not admit psc metrics.

Theorem 3.2 (GL1980, GL1982, CS2020). *Enlargeable manifolds do not admit metrics of positive scalar curvature.*

This is the result that motivated the original definition by Gromov and Lawson [GL1980], where they first showed that enlargeability is an obstruction to the existence of psc metrics, under the additional assumption that the coverings $\hat{M}$ in Definition 3.1 are finite and spin. They were later able to drop the finiteness assumption [GL1982]. Both proofs are based on techniques using the spin Dirac operator, so it was unclear how and if these results carry over to the non-spin case. Rather recently however, Cecchini and Schick resolved this issue in [CS2020] by relying on modern minimal surface methods developed by Schoen and Yau [SY...].

The following Lemma now provides two easy ways to construct new enlargeable manifolds out of given ones.

Lemma 3.3. (i) *The product of enlargeable manifolds is enlargeable.*

(ii) *A closed oriented manifold admitting a non-zero degree map onto an enlargeable manifold is enlargeable.*

Proof. For ?? let M^m and N^n be enlargeable. For $\varepsilon > 0$ choose Riemannian coverings $p: \hat{M} \rightarrow M$ and $q: \hat{N} \rightarrow N$ as well as ε -contracting maps $f: \hat{M} \rightarrow S^m$ and $g: \hat{N} \rightarrow S^n$ of non-zero degree. Then

$$p \times q: \hat{M} \times \hat{N} \rightarrow M \times N$$

is a Riemannian covering, where both manifolds are equipped with the corresponding product metric. Moreover

$$f \times g: \hat{M} \times \hat{N} \rightarrow S^m \times S^n$$

defines an ε -contracting map of degree $\deg(f \times g) = \deg(f)\deg(g) \neq 0$. Now consider the map

$$\phi: S^m \times S^n \rightarrow S^m \wedge S^n \cong S^{m+n},$$

given by projection onto the smash product

$$S^m \wedge S^n := S^m \times S^n / (S^m \times * \cup * \times S^n),$$

which is well known to be homeomorphic to the sphere S^{m+n} . The map ϕ is easily seen to be of degree ± 1 (depending on the chosen orientations) and after a slight perturbation we may also assume it to be smooth. Moreover, because $S^m \times S^n$ is closed, it follows that ϕ is a λ -Lipschitz

map for some constant $\lambda > 0$. Therefore $\phi \circ (f \times g)$ is a $\lambda\varepsilon$ -Lipschitz map of non-zero degree and since ε was chosen arbitrarily the statement follows.

For ?? let M^m be enlargeable and let $h: P \rightarrow M$ be some non-zero degree map. Again given $\varepsilon > 0$ choose a Riemannian covering $p: \hat{M} \rightarrow M$ as well as an ε -contracting map $f: \hat{M} \rightarrow S^m$. Define a covering $\pi: \hat{P} \rightarrow P$ as the pullback covering of p under h , that is π fits into a pullback diagram of the form

$$\begin{array}{ccc} \hat{P} & \xrightarrow{H} & \hat{M} \\ \downarrow \pi & & \downarrow p \\ P & \xrightarrow{h} & M \end{array}$$

where H is some suitable smooth map. Since $\hat{M}$ and P are closed, the same holds for $\hat{P}$. Moreover h is μ -Lipschitz for some constant $\mu > 0$, because P is closed. Consequently H is also μ -Lipschitz, given that both π and p are local isometries. For the degree of H we compute

$$\deg(p)\deg(H) = \deg(p \circ H) = \deg(h \circ \pi) = \deg(h)\deg(\pi) \neq 0,$$

using the fact that $\deg(h) \neq 0$ by assumption and that the degree of a covering is given by the cardinality of the fiber. We conclude that H is of non-zero degree, such that $f \circ H: \hat{P} \rightarrow S^m$ is a $\mu\varepsilon$ -Lipschitz map of non-zero degree. Again since ε was chosen arbitrarily the statement follows. $\square$

From Part ?? of the above lemma it follows immediately, that enlargeability is a homotopy invariant, since a homotopy equivalence of connected closed oriented manifolds has degree ± 1 . In fact much more is true, namely that enlargeability is a homological invariant in the following sense.

Theorem ([BH10]). *Let M be a closed oriented n -manifold and let $\phi: M \rightarrow B\pi_1(M)$ be the classifying map of the universal cover. Then M is enlargeable if and only if $\phi_*[M] \in H_n(B\pi_1(M); \mathbb{Q})$ does not lie in the rational vector subspace*

$$H_n^{\text{small}}(B\pi_1(M); \mathbb{Q}) \subseteq H_n(B\pi_1(M); \mathbb{Q})$$

of “small” classes, i.e. $\phi_*[M]$ is “large”.

In [BH10] the authors showed this by giving a general definition of “small” and “large” homology classes of a simplicial complex X with finitely generated fundamental group, such that the assignment $X \mapsto H_n^{\text{small}}(X; \mathbb{Q})$ is functorial for continuous maps $f: X \rightarrow Y$ inducing an epimorphism on π_1 , that is $f_*: H_n(X; \mathbb{Q}) \rightarrow H_n(Y; \mathbb{Q})$ maps small classes to small classes. Moreover if f happen to induce an isomorphism on π_1 then f_* also maps large classes to large classes. This general definition then specializes to the fact that a closed oriented manifold M is enlargeable if and only if the fundamental class $[M]$ is large, that is $[M] \notin H_n^{\text{small}}(M; \mathbb{Q})$. Combining these observations, the theorem follows. Furthermore this argument shows that the notion of enlargeability can be purely thought of as a concept of simplicial complexes. For details we refer the reader to [BH10].

4 Construction of the Circle Bundles

We will now construct counterexamples to the question raised in the beginning, that is circle bundles $E \rightarrow M$ over enlargeable M , where the total space E admits a metric of positive scalar curvature.

For $n \geq 2$ let N^{2n} be any $2n$ -dimensional totally non-spin manifold. We could for example take $N := \mathbb{C}P^2 \times T^{2n-4}$, which is totally non-spin by Lemma 2.7 and Lemma 2.9. Now consider the manifold

$$M := \mathbb{C}P^n \# T^{2n} \# N.$$

Since the collapse map $M \rightarrow T^{2n}$ has degree 1 and T^{2n} is enlargeable, it follows from Lemma 3.3 that M is also enlargeable. We will now construct a circle bundle $E \rightarrow M$ as follows. Let

$$c_1 \in H^2(M; \mathbb{Z}) \cong H^2(\mathbb{C}P^n; \mathbb{Z}) \oplus H^2(T^{2n}; \mathbb{Z}) \oplus H^2(N; \mathbb{Z})$$

be the cohomology class corresponding to the first chern class $\alpha \in H^2(\mathbb{C}P^n; \mathbb{Z})$ of the canonical bundle $S^{2n+1} \rightarrow \mathbb{C}P^n$ under the above isomorphism. We then define $E \rightarrow M$ as the unique S^1 -principal bundle with first chern class equal to c_1 . Now this is quite an abstract definition, so let us develop an intuition for how this bundle looks like. We claim that E is trivial over the summand $T^{2n} \# N$ and is given by the restriction of the canonical bundle $S^{2n+1} \rightarrow \mathbb{C}P^n$ to $\mathbb{C}P^n - D^{2n} \subseteq M$ over the $\mathbb{C}P^n$ summand. To make this precise let

$$i_1: (T^{2n} \# N) - D^{2n} \rightarrow M$$

be the inclusion. Then

$$\begin{array}{ccc} H^2(M; \mathbb{Z}) & \xrightarrow{i_1^*} & H^2((T^{2n} \# N) - D^{2n}; \mathbb{Z}) \\ \cong \uparrow & & \cong \uparrow \\ H^2(\mathbb{C}P^n; \mathbb{Z}) \oplus H^2(T^{2n} \# N; \mathbb{Z}) & \xrightarrow{\text{proj.}} & H^2(T^{2n} \# N; \mathbb{Z}) \end{array}$$

commutes and therefore $i_1^*(c_1) = 0$. Again using the fact that S^1 -principal bundles are uniquely determined by their first chern class we see that i_1^*E is indeed trivial. Similarly if we denote by

$$i_2: \mathbb{C}P^n - D^{2n} \rightarrow M, \quad j: \mathbb{C}P^n - D^{2n} \rightarrow \mathbb{C}P^n$$

the inclusions, the diagram

$$\begin{array}{ccc} H^2(M; \mathbb{Z}) & \xrightarrow{i_2^*} & H^2(\mathbb{C}P^n - D^{2n}; \mathbb{Z}) \\ \cong \uparrow & & \cong \uparrow j^* \\ H^2(\mathbb{C}P^n; \mathbb{Z}) \oplus H^2(T^{2n} \# N; \mathbb{Z}) & \xrightarrow{\text{proj.}} & H^2(\mathbb{C}P^n; \mathbb{Z}) \end{array}$$

commutes and thus $i_2^*(c_1) = j^*(\alpha)$. So the first chern classes of i_2^*E and j^*S^{2n+1} agree, hence they must be isomorphic bundles.

With this intuition in mind we will now show the following two claims.

Claim 1. E is totally non-spin.

Claim 2. The projection $p: E \rightarrow M$ induces an isomorphism on π_1 .

Granting these claims for now, let us see how to conclude the existence of a positive scalar curvature metric on E . The disk bundle $DE \rightarrow M$ corresponding to E defines an oriented null-cobordism of E (see Example 1.6 iii)). Moreover, since $DE \rightarrow M$ is a homotopy equivalence it follows from Claim 1 that the inclusion $E \rightarrow DE$ is an isomorphism on π_1 . So together with Claim 2 Theorem 2.5 applies, hence E admits a psc metric.

Let us now proof Claim 1 and Claim 2.

Proof of Claim 1. Let $i: N - D^{2n} \rightarrow M$ be the inclusion. We have already established above, that E is trivial over the $T^{2n} \# N$ summand and so in particular

$$i^*E \cong (N - D^{2n}) \times S^1.$$

From Lemma 2.7 (i) and (iii) we conclude that $(N - D^{2n}) \times S^1$ is totally non-spin. Therefore E contains a totally non-spin submanifold of codimension 0 and so Lemma 2.7 (ii) implies that E is also totally non-spin. $\square$

Proof of Claim 2. We have to show that the fibers of E can be contracted. This is essentially due to the fact that E contains (the restriction of) the canonical bundle $S^{2n+1} \rightarrow \mathbb{C}P^n$ for which the fibers are contractible. To formalize this let $j: \mathbb{C}P^n - D^{2n} \rightarrow \mathbb{C}P^n$ be the inclusion. Consider the following commutative diagram, where the rows are taken from the long exact sequences of homotopy groups of the respective fibrations and the vertical maps are induced by inclusions:

$$\begin{array}{ccccc}
\pi_2(M) & \xrightarrow{\partial_1} & \pi_1(S^1) & \longrightarrow & \pi_1(E) \\
\uparrow & & \uparrow \text{id} & & \uparrow \\
\pi_2(\mathbb{C}P^n - D^{2n}) & \xrightarrow{\partial_2} & \pi_1(S^1) & \longrightarrow & \pi_1(j^*E) \\
\cong \downarrow & & \downarrow \text{id} & & \downarrow \\
\pi_2(\mathbb{C}P^n) & \xrightarrow{\partial_3} & \pi_1(S^1) & \longrightarrow & \pi_1(S^{2n+1})
\end{array}$$

The vertical map on the lower left is an isomorphism, because both $\mathbb{C}P^n$ and $\mathbb{C}P^n - D^{2n}$ are simply connected and so by the Hurewicz Theorem

$$\begin{array}{ccc}
\pi_2(\mathbb{C}P^n - D^{2n}) & \longrightarrow & \pi_2(\mathbb{C}P^n) \\
\downarrow \cong & & \downarrow \cong \\
H_2(\mathbb{C}P^n - D^{2n}; \mathbb{Z}) & \longrightarrow & H_2(\mathbb{C}P^n; \mathbb{Z})
\end{array}$$

commutes and the vertical maps are isomorphisms. Since the bottom horizontal map is an isomorphism the same holds for the top horizontal map. We now claim that ∂_1 is surjective. For this it suffices to show surjectivity of ∂_2 , which in turn is equivalent to surjectivity of ∂_3 . But this is clear by exactness of the rows and $\pi_1(S^{2n+1}) = 0$. Consequently

$$\pi_1(S^1) \rightarrow \pi_1(E)$$

is the zero map and therefore we conclude from exactness of

$$\pi_1(S^1) \rightarrow \pi_1(E) \rightarrow \pi_1(M) \rightarrow \pi_0(S^1),$$

that $\pi_1(E) \rightarrow \pi_1(M)$ is indeed an isomorphism. $\square$

The above construction now gives examples for circle bundles with positive scalar curvature over enlargeable manifolds M in all even dimensions $\dim(M) \geq 4$. To also get examples in odd dimensions let $E \rightarrow M$ be as above and let $\pi: M \times S^1 \rightarrow M$ be the projection. Then the pullback bundle $\pi^*E \cong E \times S^1$ clearly admits a metric of positive scalar curvature and by Lemma 3.3 the base space $M \times S^1$ is still enlargeable. Naturally, the question arises as to what happens in dimensions $\dim(M) \leq 3$. In this case it turns out that no circle bundle $E \rightarrow M$ admits a metric of positive scalar curvature, see [KS25, Theorem B].

A Appendix

A.1 A Short Introduction to Surgery

Surgery Theory provides a way to change the topology of a manifold in a controlled way. The basic idea is the simple observation, that a product $S^k \times S^{n-k-1}$ of spheres can be considered as the boundary of both $S^k \times D^{n-k}$ and of $D^{k+1} \times S^{n-k-1}$. This allows for the following construction. Let M^n be a smooth n -dimensional manifold and let $F: S^k \times D^{n-k} \rightarrow M$ be an embedding into the interior of M . Then we can remove the interior of the image of F resulting in a manifold

$$\tilde{M} := M - \text{int}(\text{im}(F)).$$

To $\tilde{M}$ we can attach $D^{k+1} \times S^{n-k-1}$ along the boundary component $F(S^k \times S^{n-k-1}) \subseteq \tilde{M}$ to obtain a manifold M' . More precisely we define M' as the pushout

$$\begin{array}{ccc} S^k \times S^{n-k-1} & \xrightarrow{F|_{S^k \times S^{n-k-1}}} & \tilde{M} \\ \downarrow \text{incl.} & & \downarrow \\ D^{k+1} \times S^{n-k-1} & \longrightarrow & M' \end{array}$$

Up to diffeomorphism there exists a unique smooth structure on M' , such that the topological embeddings $\tilde{M} \rightarrow M'$ and $D^{k+1} \times S^{n-k-1} \rightarrow M'$ are smooth embeddings, see [Milnor1965, Theorem 1.4]. We say that the manifold M' is obtained from M by surgery along F . Intuitively we remove a thickened k -sphere from M and glue in its place a thickened $(n-k-1)$ -sphere. Accordingly we refer to the surgery along F as a surgery of dimension k or simply a k -surgery.

A different, yet closely related concept, is the theory of cobordisms. Two closed n -dimensional manifolds M and M' are called *cobordant* if there exists a compact manifold W^{n+1} whose boundary is the disjoint union of M and M' . The manifold W is called a cobordism from M to M' . Since any closed manifold is cobordant to itself via the trivial cobordism $M \times [0, 1]$ and cobordisms can be glued together along common boundary components, the relation of cobordism defines an equivalence relation on the class of closed n -dimensional manifolds. One way to construct a new cobordism out of a given one is by means of *handle attachments*. Suppose we are given an embedding $\phi: S^k \times D^{n-k} \rightarrow \partial W$. Then we can form a new manifold W' by attaching a so called $(k+1)$ -handle $D^{k+1} \times D^{n-k}$ to W along ϕ , that is we form the pushout

$$\begin{array}{ccc} S^k \times D^{n-k} & \xrightarrow{\phi} & W \\ \downarrow \text{incl.} & & \downarrow \\ D^{k+1} \times D^{n-k} & \longrightarrow & W' \end{array}$$

We also refer to $(k+1)$ as the *index* of the handle. Again some care needs to be taken to define a smooth structure on W' , see [Milnor, Morse Theory]. It turns out that any cobordism W from M to M' can be obtained from the trivial cobordism $M \times [0, 1]$ by attaching finitely many handles [REFERENCE]. Now the connection between surgery theory and cobordisms is the following fundamental result, which can be shown by means of Morse theory.

Theorem A.1 (Milnor1965, Theorem 3.13 and Theorem 3.14 + other source!!). *Let M and M' be closed manifolds of the same dimension. Then M' is obtained from M by a surgery of dimension k if and only if there exists a cobordism W from M to M' containing only a single handle of index $k+1$, that is W is obtained from the trivial cobordism $M \times [0, 1]$ by attaching a single $(k+1)$ -handle.*

One common usecase of surgery is to modify a given manifold M in order to simplify its homotopy groups. Essentially given some embedding $f: S^k \rightarrow M$ we would like to modify the topology of M , such that f becomes contractible. We could just glue in a disk D^{k+1} along f , however this would destroy the manifold structure. So instead we “thicken” the embedded sphere by extending f to an embedding $F: S^k \times D^{n-k} \rightarrow M$, remove this thickened sphere and glue in a whole family of disks $D^{k+1} \times S^{n-k-1}$, i.e. we perform a k -surgery. This process is usually referred to as “killing” the homotopy class $[f]$ by surgery. The fundamental obstruction to performing such a surgery is the existence of the embedding F , which exists if and only if the normal bundle of f is trivial. The following Proposition now formalizes this idea of killing homotopy classes by surgery.

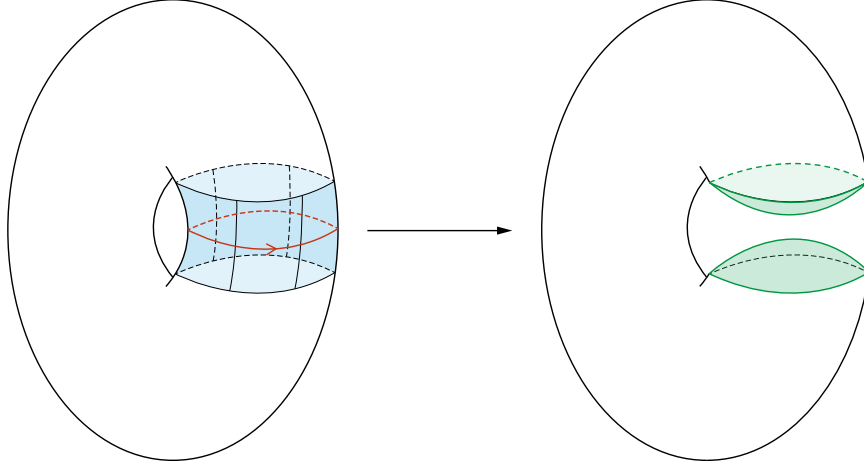


Figure 1: Test

Proposition A.2. *Let W^n be a smooth manifold with basepoint $w_0 \in W$, $M \subseteq \partial W$ a boundary component with basepoint $x_0 \in M$ and $2(k+1) \leq n$.*

- (i) *Let $\alpha_1, \dots, \alpha_l \in \pi_k(W, w_0)$ be classes, which are represented by embeddings with trivial normal bundle. Then there exists a manifold W' obtained from W by l -many k -surgeries away from the point w_0 together with an epimorphism*

$$\pi_k(W, w_0) \twoheadrightarrow \pi_k(W', w_0)$$

containing the $\mathbb{Z}[\pi_1(W, w_0)]$ -module generated by $\alpha_1, \dots, \alpha_l$ in its kernel. Moreover

$$\pi_i(W, w_0) \cong \pi_i(W', w_0)$$

for all $i \leq k-1$.

- (ii) *Let $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}} \in \pi_k(W, M, x_0)$ be classes, which lift to classes $\alpha_1, \dots, \alpha_l \in \pi_k(W, x_0)$ under the map $\pi_k(W, x_0) \rightarrow \pi_k(W, M, x_0)$ induced by the inclusion. Again assume that $\alpha_1, \dots, \alpha_l$ are represented by embeddings with trivial normal bundle. Then there exists a manifold W' obtained from W by l -many k -surgeries together with an epimorphism*

$$\pi_k(W, M, x_0) \twoheadrightarrow \pi_k(W', M, x_0)$$

containing the $\mathbb{Z}[\pi_1(M, x_0)]$ -module generated by $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ in its kernel. Moreover

$$\pi_i(W, M, x_0) \cong \pi_i(W', M, x_0)$$

for all $i \leq k-1$.

In the proof of the Proposition we will make use of the following basic lemma.

Lemma A.3. *Let X be a topological space and let $H: S^k \times [0, 1] \rightarrow X$ be a homotopy from some map $f: S^k \rightarrow X$ to some map $g: S^k \rightarrow X$. Fix a basepoint $s_0 \in S^k$ and consider the path $\gamma(t) := H(s_0, t)$ from $x_0 := f(s_0)$ to $x_1 := g(s_0)$. Then the change of basepoint isomorphism*

$$b_\gamma: \pi_k(X, x_0) \rightarrow \pi_k(X, x_1)$$

induced by γ maps the class $[f]$ to the class $[g]$.

Proof. Represent the classes $[f]$ and $[g]$ by maps $f': D^k \rightarrow X$ and $g': D^k \rightarrow X$, which are constant on the boundary. Let $H': D^k \times [0, 1] \rightarrow X$ be the corresponding homotopy from f' to g' , such that $H'(S^{k-1}, t) = \gamma(t)$. Then by $b_\gamma([f'])$ is by definition represented by

$$D^k \rightarrow X, \quad y \mapsto \begin{cases} f'(2y), & \|y\| \leq \frac{1}{2}, \\ \gamma(2\|y\| - 1), & \|y\| \geq \frac{1}{2}. \end{cases}$$

This map is homotopic to g' relative S^{k-1} by virtue of the following homotopy.

$$D^k \times [0, 1] \rightarrow X, \quad y \mapsto \begin{cases} H'\left(\frac{2}{1+t}y, t\right), & \|y\| \leq \frac{1}{2}(1+t), \\ \gamma(2\|y\| - 1), & \|y\| \geq \frac{1}{2}(1+t). \end{cases}$$

□

Proof of Proposition A.2. We first prove the absolute version ?? of the Proposition in three steps.

Step 1. We start by explaining how to kill a single homotopy class by surgery. Suppose that $w_0 \in W$ is an interior point. Let $\beta \in \pi_k(W, w_0)$ be a class represented by an embedding $f: (S^k, s_0) \rightarrow (W, w_0)$ for a specified basepoint $s_0 \in S^k$, which extends to an embedding $F: S^k \times D^{n-k} \rightarrow W$ into the interior of W . Let

$$\tilde{W} := W - \text{int}(\text{im}(F))$$

and define W' by the pushout

$$\begin{array}{ccc} S^k \times S^{n-k-1} & \longrightarrow & \tilde{W} \\ \downarrow & & \downarrow \\ D^{k+1} \times S^{n-k-1} & \longrightarrow & W' \end{array}$$

as discussed above. Since the inclusion $S^k \times S^{n-k-1} \rightarrow D^{k+1} \times S^{n-k-1}$ is a cofibration it follows that the above square is a homotopy pushout, hence the map $\tilde{W} \rightarrow W'$ has the same connectivity as $S^k \times S^{n-k-1} \rightarrow D^{k+1} \times S^{n-k-1}$, which is k -connected. Similarly, the pushout

$$\begin{array}{ccc} S^k \times S^{n-k-1} & \longrightarrow & \tilde{W} \\ \downarrow & & \downarrow \\ S^k \times D^{n-k} & \longrightarrow & W \end{array}$$

implies that $\tilde{W} \rightarrow W$ is $n - k - 1$ -connected. Choose any point $d_0 \in \partial D^{n-k} = S^{n-k-1}$ and set $w'_0 := F(s_0, d_0)$. We now construct an epimorphism $\Phi_k: \pi_k(W, w_0) \rightarrow \pi_k(W', w'_0)$ mapping the class β to 0. Choose a path $\lambda: [0, 1] \rightarrow D^{n-k}$ from the center point $0 \in D^{n-k}$ to d_0 and let

$$\eta(t) := F(s_0, \lambda(t)).$$

Then η is a path from w_0 to w'_0 and we can consider the change of basepoint isomorphism

$$b_\eta: \pi_k(W, w_0) \rightarrow \pi_k(W, w'_0)$$

induced by η . From the preceding arguments it follows, that the map $\pi_i(\tilde{W}, w'_0) \rightarrow \pi_i(W, w'_0)$ induced by the inclusion is an isomorphism for $i \leq k$, since $2(k+1) \leq n$. Thus we can define maps

$$\Phi_i: \pi_i(W, w_0) \xrightarrow{b_\eta} \pi_i(W, w'_0) \xleftarrow{\cong} \pi_i(\tilde{W}, w'_0) \rightarrow \pi_i(W', w'_0).$$

By the above discussion Φ_i is an isomorphism for $i \leq k - 1$ and an epimorphism for $i = k$. We now show that Φ_k maps the class β to 0. Define

$$\tilde{f}: (S^k, s_0) \rightarrow (W, w'_0), \quad y \mapsto F(y, d_0).$$

Then $\tilde{f}$ is freely homotopic to f via the homotopy $(y, t) \mapsto F(y, \lambda(t))$, so it follows from Lemma A.3 that $b_\eta(\beta)$ is represented by $\tilde{f}$. Note that the image of $\tilde{f}$ lies in $\tilde{W}$ and thus $\Phi_k(\beta)$ is represented by

$$(S^k, s_0) \rightarrow (W', w'_0), y \mapsto \tilde{f}(y).$$

This map factors as the following composition of inclusions

$$S^k \rightarrow D^{k+1} \times \{d_0\} \rightarrow W',$$

which is clearly contractible. Therefore $\Phi_k(\beta) = 0$. Note that all maps in the definition of Φ_k are compatible with the respective π_1 -actions, hence the entire $\mathbb{Z}[\pi_1(W, w_0)]$ -module generated by β is contained in the kernel of Φ_k .

Step 2. Next we explain how to kill multiple homotopy classes at once. To this end assume that W is connected. Let $\beta_i \in \pi_k(W, w_i)$ for interior points $w_i \in W$, $1 \leq i \leq l$. Again suppose that the β_i can be represented by embeddings $f_i: (S^k, s_0) \rightarrow (W, w_i)$, which extend to embeddings $F_i: S^k \times D^{n-k} \rightarrow W$ into the interior of W . Moreover assume these embeddings to be disjoint, that is $\text{im}(F_i) \cap \text{im}(F_j) = \emptyset$ for $i \neq j$. This allows us to do surgery along all F_i at once, i.e. we can define

$$\tilde{W} := W - \bigcup_{i=1, \dots, l} \text{int}(\text{im}(F_i))$$

and let W' be defined by the pushout

$$\begin{array}{ccc} \coprod_{i=1, \dots, l} S^k \times S^{n-k-1} & \longrightarrow & \tilde{W} \\ \downarrow & & \downarrow \\ \coprod_{i=1, \dots, l} D^{k+1} \times S^{n-k-1} & \longrightarrow & W' \end{array}$$

In order to make sense of the statement, that we can kill all homotopy classes β_i at once we would like to consider them as elements in the same group. To achieve this let $d_0 \in S^{n-k-1}$ and $\lambda: [0, 1] \rightarrow D^{n-k}$ be as in Step 1 and set

$$w'_i := F_i(s_0, d_0), \quad \eta_i(t) := F_i(s_0, \lambda(t)).$$

Moreover choose any point $v_0 \in \tilde{W}$ together with paths $\gamma'_i: [0, 1] \rightarrow \tilde{W}$ from $w'_i \rightarrow v_0$. Such paths exist, since W was assumed to be connected and removing (thickened) spheres of codimension at least 2 preserves connectedness. Let

$$\gamma_i := \gamma'_i * \eta_i: [0, 1] \rightarrow W$$

denote the composed path from w_i to v_0 and consider the following commutative diagram.

$$\begin{array}{ccccccc} \pi_k(W, v_0) & \xrightarrow{\text{id}} & \pi_k(W, v_0) & \xleftarrow{\cong} & \pi_k(\tilde{W}, v_0) & \longrightarrow & \pi_k(W', v_0) \\ b_{\gamma_i} \uparrow \cong & & b_{\gamma'_i} \uparrow \cong & & b_{\gamma'_i} \uparrow \cong & & b_{\gamma'_i} \uparrow \cong \\ \pi_k(W, w_i) & \xrightarrow[b_{\eta_i}]{\cong} & \pi_k(W, w'_i) & \xleftarrow{\cong} & \pi_k(\tilde{W}, w'_i) & \longrightarrow & \pi_k(W', w'_i) \end{array}$$

As we have seen in Step 1, the class β_i is contained in the kernel of the bottom horizontal composition, hence $b_{\gamma_i}(\beta_i)$ is contained in the kernel of the top horizontal composition, which defines a map

$$\pi_k(W, v_0) \rightarrow \pi_k(W', v_0).$$

As in Step 1 we conclude that this map is an epimorphism containing the $\mathbb{Z}[\pi_1(W, v_0)]$ -module generated by $b_{\gamma_1}(\beta_1), \dots, b_{\gamma_l}(\beta_l)$ in its kernel. Moreover, the same arguments as in Step 1 apply to show that $\pi_j(W, v_0) \cong \pi_j(W', v_0)$ for $j \leq k-1$.

Step 3. Finally we show how to kill multiple homotopy classes as formulated in ???. Let $g_i: (S^k, s_0) \rightarrow (W, w_0)$ be embeddings with trivial normal bundle representing the class α_i . By a general position argument we can find isotopies $H_i: S^k \times [0, 1] \rightarrow W$ from g_i to embeddings $f_i: S^k \rightarrow W - \{w_0\}$, such that $f_i(S^k) \cap f_j(S^k) = \emptyset$ for $i \neq j$. Moreover we may assume, that the

image of f_i is contained in the interior of W . Since the normal bundle of g_i is trivial, the same holds for f_i . Therefore we can extend the f_i to embeddings

$$F_i: S^k \times D^{n-k} \rightarrow W - \{w_0\}$$

into the interior of W , such that $\text{im}(F_i) \cap \text{im}(F_j) = \emptyset$ for $i \neq j$. Now we are in the setting as in Step 2 for $w_i = f_i(s_0)$ and $\beta_i = [f_i] \in \pi_k(W, w_i)$. Define paths $\gamma_i(t) := H_i(s_0, 1-t)$ from w_i to w_0 . Then $b_{\gamma_i}(\beta_i) = \alpha_i$ by Lemma A.3. We would like to apply Step 2 to $v_0 = w_0$, however the paths γ_i are not of the same form as in Step 2 yet. Let us ignore this issue for a moment and see how Step 3 follows. Again let W' be the manifold obtained by surgery along $F_1, \dots, F_l$ as before. Step 2 now yields an epimorphism

$$\Phi: \pi_k(W, w_0) \rightarrow \pi_k(W', w_0).$$

containing the $\mathbb{Z}[\pi_1(W, w_0)]$ -module generated by the $b_{\gamma_i}(\beta_i) = \alpha_i$ in its kernel. Moreover $\pi_j(W, w_0) \cong \pi_j(W', w_0)$ for $j \leq k-1$.

We now construct a homotopy to bring γ_i into the desired form. Let $t_i := \sup\{t \in [0, 1] \mid \gamma_i([0, t]) \subseteq \text{im}(F_i)\}$ be the first time at which γ_i exits $\text{im}(F_i)$. For $t \in [0, t_i]$ we can then write $\gamma_i(t)$ in the form

$$\gamma_i(t) = F_i(\tilde{\sigma}_i(t), \tilde{\lambda}_i(t)).$$

for suitable paths $\tilde{\sigma}_i, \tilde{\lambda}_i$. Now $\tilde{\sigma}_i$ and $\tilde{\lambda}_i$ are homotopic to the reparametrized paths

$$\sigma_i(t) = \begin{cases} \tilde{\sigma}_i(0) = s_0, & t \in \left[0, \frac{t_i}{2}\right] \\ \tilde{\sigma}_i(2t - t_i), & t \in \left[\frac{t_i}{2}, t_i\right] \end{cases} \text{ and } \lambda_i(t) = \begin{cases} \tilde{\lambda}_i(2t), & t \in \left[0, \frac{t_i}{2}\right] \\ \tilde{\lambda}_i(t_i), & t \in \left[\frac{t_i}{2}, t_i\right] \end{cases}$$

respectively. So after applying this homotopy we may replace $\tilde{\sigma}_i$ and $\tilde{\lambda}_i$ by σ_i and λ_i . For $t \in [0, \frac{t_i}{2}]$, $\gamma_i(t)$ can then be written as

$$\gamma_i(t) = F_i(s_0, \lambda_i(t)),$$

which is of the same form as η_i in Step 2. It remains to bring $\gamma'_i := \gamma_i|_{[\frac{t_i}{2}, 1]}$ into the desired form, meaning the image of γ'_i should be completely contained within $\tilde{W}$. We can achieve this as follows. Applying a general position argument we can assume, that γ'_i does not intersect any of the embedded spheres $f_1(S^k), \dots, f_l(S^k)$. Consider the deformation retraction

$$r: W - \bigcup_{j=1, \dots, l} f_j(S^k) \rightarrow \tilde{W}$$

induced by the deformation retractions $S^k \times (D^{n-k} - 0) \rightarrow S^k \times S^{n-k-1}$. Then $r \circ \gamma'_i$ is a path in $\tilde{W}$, which is homotopic to γ'_i relative to the endpoints. So in conclusion we may indeed assume, that γ_i is of the form as in Step 2.

Step 4. We can now deduce the relative version ?? of the proposition using ?? applied to the given classes $\alpha_1, \dots, \alpha_l \in \pi_k(W, x_0)$. Using these classes construct $\tilde{W}$ and W' as before. For $i \leq k$ consider the following diagram, where all maps are induced by inclusions and the rows are exact.

$$\begin{array}{ccccccccc} \pi_i(M, x_0) & \longrightarrow & \pi_i(W, x_0) & \longrightarrow & \pi_i(W, M, x_0) & \longrightarrow & \pi_{i-1}(M, x_0) & \longrightarrow & \pi_{i-1}(W, x_0) \\ \uparrow \text{id} & & \uparrow \cong & & \uparrow & & \uparrow \text{id} & & \uparrow \cong \\ \pi_i(M, x_0) & \longrightarrow & \pi_i(\tilde{W}, x_0) & \longrightarrow & \pi_i(\tilde{W}, M, x_0) & \longrightarrow & \pi_{i-1}(M, x_0) & \longrightarrow & \pi_{i-1}(\tilde{W}, x_0) \\ \downarrow \text{id} & & \downarrow & & \downarrow & & \downarrow \text{id} & & \downarrow \cong \\ \pi_i(M, x_0) & \longrightarrow & \pi_i(W', x_0) & \longrightarrow & \pi_i(W', M, x_0) & \longrightarrow & \pi_{i-1}(M, x_0) & \longrightarrow & \pi_{i-1}(W', x_0) \end{array}$$

It follows from the Five Lemma that

$$\pi_i(\tilde{W}, M, x_0) \rightarrow \pi_i(W, M, x_0)$$

is an isomorphism and

$$\pi_i(\tilde{W}, M, x_0) \rightarrow \pi_i(W', M, x_0)$$

is an isomorphism for $i \leq k - 1$. This shows $\pi_i(W, M, x_0) \cong \pi_i(W', M, x_0)$ for $i \leq k - 1$. If $i = k$ the map $\pi_k(\tilde{W}, x_0) \rightarrow \pi_k(W', x_0)$ is surjective and a diagram chase shows that

$$\pi_k(\tilde{W}, M, x_0) \rightarrow \pi_k(W', M, x_0)$$

is surjective as well. Thus we can define the epimorphism

$$\Phi_{\text{rel}}: \pi_k(W, M, x_0) \xleftarrow{\cong} \pi_k(\tilde{W}, M, x_0) \rightarrow \pi_k(W', M, x_0).$$

We have already seen that $\alpha_1, \dots, \alpha_l$ are contained in the kernel of the map

$$\pi_k(W, x_0) \xleftarrow{\cong} \pi_k(\tilde{W}, x_0) \rightarrow \pi_k(W', x_0).$$

So commutativity of the above diagram implies

$$\Phi_{\text{rel}}(\alpha_i^{\text{rel}}) = 0.$$

Again all maps in the definition of Φ_{rel} are compatible with the respective π_1 -actions, such that Φ_{rel} contains the entire $\mathbb{Z}[\pi_1(M, x_0)]$ -module generated by $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ in its kernel. $\square$

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